

Article 705: Interconnected Electric Power Production Sources

by Brian Mehalic

Article 705 of the *National Electrical Code* (NEC), “Interconnected Electric Power Production Sources,” has expanded and contains some significant changes. This Code Article applies to any power-production system connected to the utility through an inverter, regardless of the energy source—examples include generators, PV systems, wind turbines, and fuel cells. Some general requirements are that interactive systems must:

- Use interactive and/or multimode inverters “listed and identified for interconnection service” (705.4)
- Be compatible with the electrical characteristics (voltage, wave shape, and frequency) of the primary power source, meaning the utility (705.14)
- Address all potential fault currents (705.16)
- Interconnect on the supply (line) side of ground-fault protection equipment (705.32)
- Disconnect upon loss of the primary power supply (705.40). This includes disconnecting all ungrounded conductors when one phase in a three-phase system is lost (705.42).

Article 705 also stipulates that all power sources on the premises need to be marked per the requirements of Sections 705.10 and 705.12(D)(3):

- At the service equipment (in the case of the utility);
- At any interconnected power production sources (the “point of connection”)
- When equipment contains multiple sources of supply (such as utility-supplied breakers and a back-fed breaker from a PV system).

Article 705 applies to any interconnected source, but its evolution has paralleled the development of Article 690 from the tremendous growth in PV installations—so some content is duplicated. These include equipment disconnecting means (690.15 and 705.21), circuit sizing (690.8 and 705.60), overcurrent protection (690.9 and 705.65), and grid-tied inverters in non-readily accessible locations [690.15(A) and 705.70].

New in Section 705.2 is the inclusion of “multimode inverters,” which can operate as both stand-alone and grid-tied inverters (examples include OutBack Power’s Radian series and Schneider Electric’s Conext XW inverters). PV systems with multimode inverters are subject to the requirements of Articles 705 and 690, including Section 690.10, “Stand-

Alone Systems” (see “Additional Fasteners,” below). Section 705.80 pertains to interactive systems with energy storage, and dictates that the maximum operating voltage (including any equalization voltage) and the polarity of the grounded conductor be marked.

Load-Side Connections

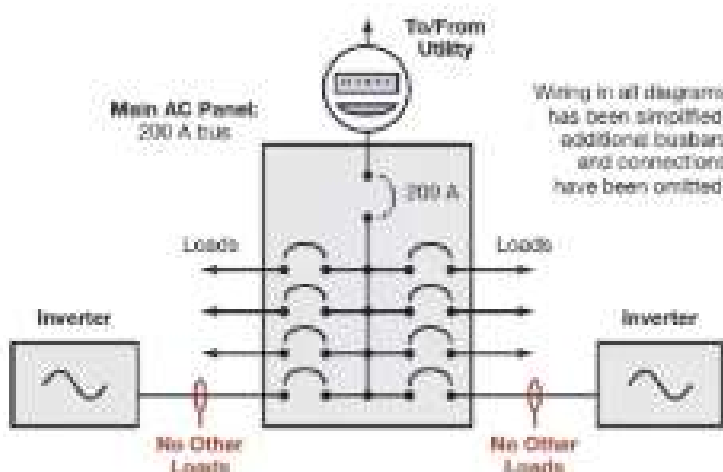
The 2014 NEC now addresses feeders, taps, and busbars in systems that are interconnected on the load side of the main AC service. Also new is the requirement to use 125% of the inverter output circuit current, rather than the size of the overcurrent protection device (OCPD) in the circuit, for determining the required busbar and conductor ampacity ratings.

Dedicated OCPD & Disconnect

The language in 705.12(D)(1) referring to “one or more inverters installed in one system” means that the interconnection circuit be dedicated to the PV system up to the disconnecting means and OCPD required by 705.20 and 705.65.

Defined in 705.2 as a “utility-interactive inverter output circuit,” these conductors—between the inverter and its interconnection OCPD—cannot supply power to any other loads, nor can they extend past the inverter to supply loads. They are strictly dedicated to the connection of an electric power production source to the primary source. This point of connection can occur at “any distribution equipment on the premises,” such as a subpanel. There also can be multiple PV systems interconnected through their own separate, dedicated means.

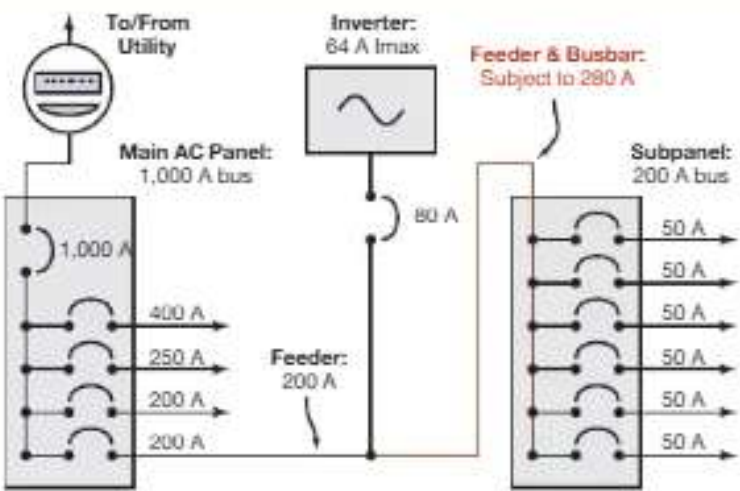
Dedicated OCPD & Disconnect



Feeders

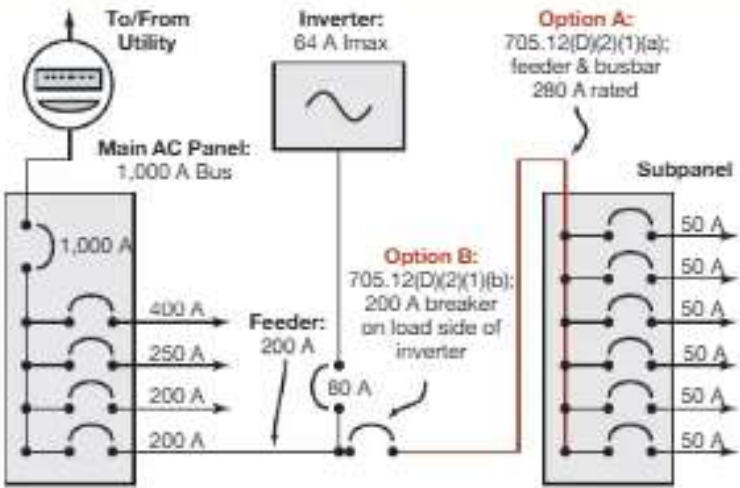
Section 705.12(D)(2)(1) addresses protecting feeder conductors, which are defined in Article 100 as “conductors between the service equipment...or other power supply source and the final branch-circuit overcurrent device.” A common example would be the conductors between the main service panel and a subpanel in a home or building. The concern is that if a PV system is connected to a feeder at some point other than its opposite end, some section of the feeder could be subject to higher, additive currents: from the primary source OCPD supplying the feeder and from the inverter output.

Feeder Overload Concern



There are two methods for protecting the portion of the feeder which is subject to the combined output of the supply breaker and the inverter. One option is to add another OCPD—rated to protect the feeder conductor and typically the same size as the feeder’s existing primary supply breaker—on the load side of the inverter connection to the feeder. In the other method, the feeder ampacity (as well as the busbar rating, if

Feeder Protection Options:
705.12(D)(2)(1)(a) & (b)



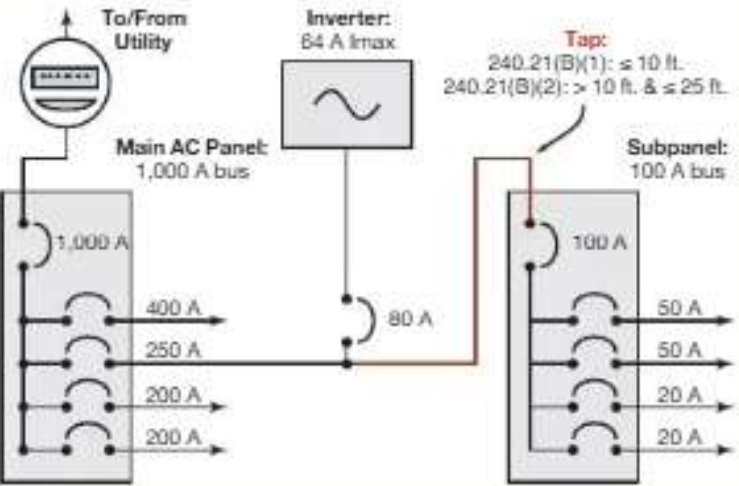
there is a main lug-only panel) can be increased so that it has an ampacity greater than or equal to the primary supply breaker plus 125% of the inverter output current.

Taps

Tap conductors, defined in Article 240, “Overcurrent Protection,” are addressed in Section 705.12(D)(2)(2). Essentially, a tap is a conductor, smaller than the feeder conductor, that is connected to the feeder without overcurrent protection at that point of supply; as such, it is subject to current that could exceed its ampacity. The tap conductor then terminates in a piece of distribution equipment (such as a subpanel) that contains a main breaker. Sizing requirements for tap conductors, which are very common in commercial buildings, are found in Sections 240.21(B)(1) and (2).

Like protecting the feeder conductor, the concern is the addition of a second source of current on the feeder to which the tap is connected. Provided the numerous requirements of Section 240.21(B) are met, a tap up to 10 feet long must have an ampacity equal to or greater than 10% of the feeder supply breaker’s ampacity, and equal to or greater than the ampacity of the OCPD at the tap conductor’s terminal end. For taps between 10 and 25 feet, the ampacity has to be at least one-third (or 33.3%) of the feeder supply breaker’s ampacity (and greater than or equal to the terminal OCPD). When a PV system is connected to the feeder supplying the tap, Section 705.12(D)(2)(2) requires that 125% of the inverter’s output current must also be included in the calculation for the tap conductor ampacity.

Tap Conductor Sizing: 705.12(D)(2)(2)



Tap Ampacity Calculation:*

Taps ≤ 10 ft: (125% inverter Imax + Feeder OCPD) x 0.10
(80 A + 250 A) x 0.10 = 33 A

Minimum tap conductor ampacity = 100 A due to 100 A terminal OCPD in subpanel

Taps > 10 ft and ≤ 25 ft: (125% inverter Imax + Feeder OCPD) x 0.33
(80 A + 250 A) x 0.33 = 109 A

Minimum tap conductor ampacity = 109 A

*Provided 240.21(B) requirements are met

Supply-Side Connections

Section 705.12(A), in conjunction with the allowance in 230.82(6), permits power production systems to be connected on the supply side of the main service disconnect, typically between the meter and the premise's main breaker. The system size that can be connected on the supply side is limited by the size of the existing service. For example, a building with a 200 amp (A) service could have a PV system connected on the supply side through a 200 A fused disconnect, which computes to a maximum inverter output current of 160 A ($160 \text{ A} \times 1.25 = 200 \text{ A}$).

Section 705.31, new to the 2014 NEC, stipulates that overcurrent protection for supply-side interconnections must be located within 10 feet of the point of interconnection, unless cable limiters (devices that isolate conductors from short circuits, but don't necessarily provide overcurrent protection) or current-limited circuit breakers are installed. This helps reduce the risk from fault current sourced from the primary electricity source (the utility). For general information on supply-side connections, see "Code Corner" in HP150.

Additional Fasteners

In situations using back-fed breakers, additional fasteners may be required to reduce shock risk associated with pulling an energized breaker off a busbar. Section 408.36(D) in the 2014 NEC dictates the requirements for additional fasteners on back-fed breakers. However, Section 705.12(D)(5) states that plug-in-type back-fed breakers used for load-side connection to the utility, and connected to listed grid-tied inverters, are not subject to the additional fastener requirements. Of course back-fed breakers have to be suitable for the application, i.e., not marked "line" and "load."

Per 690.10(E), these same plug-in-type breakers, when connected to a stand-alone inverter or the stand-alone output of a multimode inverter, are required to have an additional fastener. The logic is that the back-fed breaker terminals or tabs could still be energized when pulled off the busbar, posing a shock hazard—and an additional fastener provides another level of protection. This is not the case with the output circuit of a grid-tied inverter—if the breaker is pulled off the busbar, the inverter will immediately cease operation. A stand-alone inverter's output, regardless of inverter type, would stay energized—much like a breaker connected to a fossil fuel generator, and thus is required to have the additional fastener.



This back-fed breaker has been secured with an additional fastener, providing another level of protection against shock hazards.

The required fasteners are available from most major equipment manufacturers. Another option is to connect the breakers to a main breaker that is fastened in place in the breaker panel, rather than back-feeding a plug-in breaker.

Other Considerations

In many grid-tied inverters, only the ungrounded conductors carry AC output current. The AC grounded (neutral) conductor is used as a reference for "instrumentation, voltage detection, or phase detection." In this case, per 705.95(B), the neutral conductor in the inverter output circuit only has to be as large as the equipment-grounding conductor (which is typically smaller than the phase conductors). Especially on larger inverters, this can result in cost savings, as smaller-gauge wire can be used, but be sure to verify this with the inverter manufacturer.

New in 2014 is 705.12(D)(6), which requires AC arc-fault protection for grid-tied inverters, with wiring harness (i.e., "AC interconnect cable" or "trunk cabling") output circuits rated at 240 VAC and 30 A or less, unless the harnesses are installed within an enclosed raceway. This provides added protection for microinverter and AC module systems, which often have more cabling under modules than string inverter systems. This may be difficult to implement given a lack of suitable equipment for the purpose—most AC AFCI breakers are not tested for back-feed. Some jurisdictions may waive this requirement, citing the allowance in Section 90.4 for products that are required but not yet available.

web extra

Busbars. The significant changes regarding calculations for busbar ratings in load-side interconnections are detailed in 705.12(D)(2)(3)(a) through (d). For more information, see "Code Corner" in HP160 at homepower.com/160.84.

